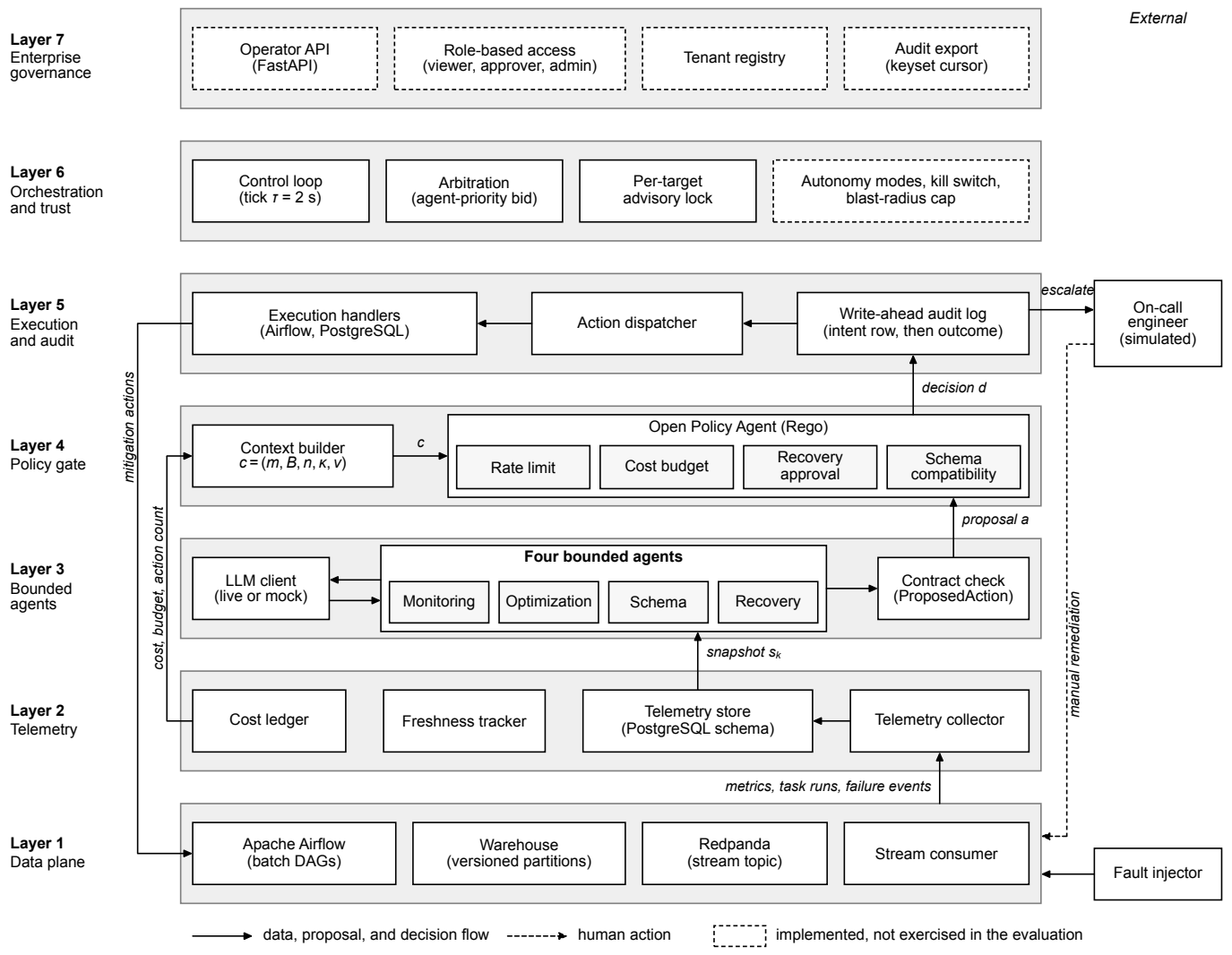

External
Layer 7
Enterprise governance
Operator API (FastAPI)
Role-based access (viewer, approver, admin)
Tenant registry
Audit export (keyset cursor)
Layer 6
Orchestration and trust
Control loop (tick τ = 2 s)
Arbitration (agent-priority bid)
Per-target advisory lock
Autonomy modes, kill switch, blast-radius cap
Layer 5
Execution and audit
Execution handlers (Airflow, PostgreSQL)
Action dispatcher
Write-ahead audit log (intent row, then outcome)
escalate
On-call engineer (simulated)
decision d
Layer 4
Policy gate
Context builder $c = (m, B, n, \kappa, v)$
c
Open Policy Agent (Rego)
Rate limit
Cost budget
Recovery approval
Schema compatibility
proposal a
Layer 3
Bounded agents
LLM client (live or mock)
Four bounded agents
Monitoring
Optimization
Schema
Recovery
Contract check (ProposedAction)
snapshot s_k
Layer 2
Telemetry
Cost ledger
Freshness tracker
Telemetry store (PostgreSQL schema)
Telemetry collector
metrics, task runs, failure events
Layer 1
Data plane
Apache Airflow (batch DAGs)
Warehouse (versioned partitions)
Redpanda (stream topic)
Stream consumer
Fault injector
mitigation actions
cost, budget, action count
manual remediation
data, proposal, and decision flow
human action
implemented, not exercised in the evaluation